

Ayame: Fast Logging for Serial Safety Net

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Abstract: Concurrency control is essential for database access, and the Serial Safety Net (SSN) represents one of the state-of-the-art methods in this field. Integrating SSN with logging ensures that transaction processing is recoverable. While P-WAL is an efficient logging method that leverages parallel I/O extensively, it still suffers from certain limitations. To overcome these drawbacks, this paper proposes a novel logging method, Ayame. The proposed method incorporates three optimization techniques: the elimination of the separate WAL-side global LSN allocation, dependency analysis, and an asynchronous durable-acknowledgment pipeline with background batched flushing. Experimental results demonstrate that Ayame achieves a 4.1- to 6.5-fold performance improvement over P-WAL in YCSB-A.

1. Introduction

1.1 Motivation

Transaction processing is a core abstraction behind data-intensive applications such as banking, payment processing, inventory management, ticket reservation, and on-line services that update shared state under high concurrency. A transaction processing system must provide both correct concurrent execution and crash recovery; in this sense, transaction processing consists of concurrency control (CC) and recovery [1]. A large body of work has studied scalable CC for multicore main-memory databases, including Silo, ERMIA, TicToc, SI, SSI, and SSN-style protocols [2], [3], [4], [5], [6], [7]. These protocols improve isolation and scalability by reducing centralized validation, timestamp, and version-management bottlenecks. In contrast, the recovery side of modern CC protocols has been explored less systematically: when a high-performance CC protocol is combined with crash durability, it is often unclear which recovery protocol preserves both safety and scalability.

Write-ahead logging (WAL) is the standard foundation for database recovery, with ARIES being the classical design [8], [9]. Modern systems have revisited WAL for multicore and fast-storage environments. SiloR parallelizes logging, checkpointing, and recovery for Silo; Passive Group Commit reduces synchronization overhead on non-volatile memory; and P-WAL partitions the log into multiple streams [10], [11], [12]. Although these systems differ in the target storage device, log format, and recovery proce-

dures, they share an important lesson: writing log records in parallel is essential for making durability compatible with multicore transaction processing.

At the same time, strong multiversion protocols such as SI, SSI, and SSN introduce timestamp and dependency structures that are not fully reflected in the parallel logging designs described above. The motivation of this work is therefore to clarify how parallel logging systems should be combined with such timestamp-based CC protocols, especially SSN, without unnecessarily reintroducing global ordering and durable-prefix bottlenecks.

1.2 Problem

We target SSN because it is the most complex of the SI-family protocols we consider: it certifies serializability using dependency metadata, while the underlying engine still uses timestamped multiversion execution. As the baseline recovery protocol, we start from P-WAL [12], which is also used as a foundation of Shirakami, the transaction processing mechanism of Tsurugi relational database system[?]. P-WAL removes the single-WAL insertion bottleneck by assigning transactions to multiple log streams. However, combining P-WAL with SSN exposes three problems.

P1. Multiple Global Counter Accesses: First, P-WAL can introduce frequent a centralized global-counter accesses. Even if the CC layer already assigns a commit timestamp for SSN validation and serialization, the recovery layer may allocate a separate global LSN or maintain a global durable prefix. This duplicates the ordering work already performed by the CC layer and adds shared-memory synchronization to the commit path.

P2. Waiting for Data Transfer: Second, workers can spend a long time waiting around `fdatasync`. P-WAL parallelizes log streams, but a transaction cannot be acknowledged until the corresponding durable condition is satisfied.

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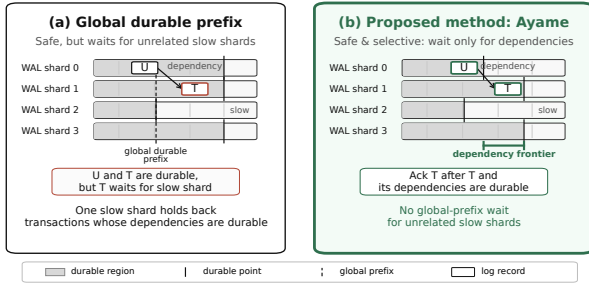


Fig. 1 Durable acknowledgment policies in parallel WAL. Global-prefix acknowledgment is safe but can delay transactions whose dependencies are already durable because it waits for unrelated slow shards. Local-only acknowledgment avoids this delay but is unsafe. Ayame uses dependency-closed acknowledgment, acknowledging a transaction only after its own log record and dependency frontier are durable.

If worker threads wait directly for log flushes or durable-prefix updates, storage synchronization and notification latency reduce the effective scalability of the CC protocol.

P3. Requiring Total History: Third, global-prefix acknowledgment waits for log records that are unrelated to the transaction being acknowledged. This rule is safe, but overly conservative: if one WAL shard is slow, transactions on other shards may be delayed even when they have no dependency on transactions logged by the slow shard. The opposite design point, local-only acknowledgment, avoids this delay but is unsafe. If transaction T reads a value written by transaction U , acknowledging T before U is recoverable may allow a crash recovery state that contains T without the state on which T depends.

Figure 1 summarizes these alternatives. Global-prefix acknowledgment is safe but may wait for unrelated slow shards. Local-only acknowledgment avoids such waits but can acknowledge a transaction before its dependencies are recoverable. The desired point is dependency-closed acknowledgment: acknowledge a transaction after its own log record, and the transactions it depends on are durable.

In asynchronous durability protocols, logical commits and durable commit acknowledgments must be distinguished. A worker may continue after enqueueing a log record, but the transaction is not durable from the client's perspective until the system returns a durable acknowledgment. Therefore, this paper uses *durable-acknowledgment throughput* as the main throughput metric and reports it together with pending durable commits and tail latency.

1.3 Approach

This paper proposes *Ayame*, a timestamp-integrated, dependency-closed parallel WAL protocol for main-memory transaction processing. Ayame targets timestamp-based engines such as ERMIA, where the transaction engine already computes a logical commit order. Its goal is not merely to maximize raw logical commit throughput, but to return durable commit acknowledgments safely without unnecessary global-prefix waiting. Ayame addresses the three problems above with three corresponding techniques.

A1. Commit timestamps as logical LSNs. Ayame reuses the SSN/ERMIA commit timestamp as the logical LSN for recovery. WAL shards still maintain local physical positions, but the WAL layer does not allocate an additional global logical order. This removes duplicated global ordering between CC and recovery and eliminates WAL-side separate global LSN allocation on the commit path.

A2. Worker/flusher/commmitter separation. Ayame decouples transaction execution from durable acknowledgment. Workers execute and validate transactions, assign commit timestamps, and enqueue log records. Flushers batch and persist shard-local logs. Committers observe durable-frontier advances and return durable acknowledgments. This pipeline keeps workers from directly blocking on `fdatasync` or on durable-prefix notification waits.

A3. Dependency-closed durable acknowledgment. Ayame replaces global-prefix acknowledgment with dependency-closed acknowledgment. For each transaction, Ayame maintains a dependency frontier that summarizes the log positions that must be recoverable before the transaction can be acknowledged. A transaction is acknowledged only when its own log record and this dependency frontier are durable. This preserves recovery safety while avoiding waits for unrelated WAL shards.

We implement Ayame in an ERMIA-style engine and evaluate it against P-WAL using SSN and YCSB workloads. The detailed experimental methodology, figures, and tables are presented in Section 4.

1.4 Organization

The rest of this paper is organized as follows. Section 2 describes transaction processing, including serial-safety-net and parallel logging protocols. Section 3 presents Ayame, which is a fast logging protocol. Section 4 describes the evaluation of Ayame compared with the P-WAL protocol using SSN. Section 5 discusses related work. Section 6 finally concludes this paper.

2. Preliminaries

2.1 Concurrency Control

A database transaction is a unit of execution that must satisfy atomicity, consistency, isolation, and durability [8]. Concurrency control (CC) is responsible for isolation: it interleaves the operations of concurrent transactions so that the outcome is equivalent to some serial execution, the standard correctness criterion of serializability [1]. CC protocols differ in how they detect and resolve the read-write, write-read, and write-write conflicts that would otherwise violate serializability. Pessimistic protocols such as two-phase locking block conflicting operations as they occur, whereas optimistic protocols let transactions execute first and validate them at commit time, aborting those whose interleaving cannot be serialized. Main-memory multicore databases favor optimistic and multiversion designs because they avoid the lock-manager and latch contention that limit scalability on many cores [2], [3], [7].

2.2 Timestamp-based Protocols

Multiversion concurrency control retains several versions of each record so that reads need not block writes. Snapshot isolation (SI) runs each transaction against a consistent snapshot taken at its start and, at commit, checks only for write–write conflicts; it is efficient but admits write-skew anomalies and is therefore not serializable. Serializable Snapshot Isolation (SSI) strengthens SI by detecting the dangerous read–write antidependency structures that can lead to non-serializable schedules [5]. These protocols order transactions with timestamps: a transaction reads as of a begin timestamp and commits at a commit timestamp, and the commit timestamps induce a logical version order. For serializable executions this version order is also a serialization order, and a recovery protocol can reuse it as the order in which to replay committed transactions after a crash.

2.2.1 Serial Safety Net

The Serial Safety Net (SSN) is a certifier layered on top of a multiversion engine such as SI or ERMIA to guarantee serializability [3], [6]. Rather than maintaining the full dependency graph, SSN summarizes, for each transaction T , the commit timestamps of its predecessors and successors in the dependency graph into a pair of watermarks, and it refuses to commit T when these watermarks reveal that committing T would close a non-serializable cycle. Each committing transaction receives a commit timestamp (*cstamp*) that fixes its position in the serialization order, and the read and write footprints of in-flight and recently committed transactions are tracked through a shared transaction-mapping table so that validation can read the relevant predecessors’ and successors’ timestamps. SSN admits more concurrency than two-phase locking while remaining serializable, which makes it a strong but demanding target for a recovery protocol. The recovery layer must respect the same commit-timestamp order that SSN uses for serialization, yet it must do so without introducing new centralized bottlenecks of its own. The shared commit-timestamp counter and transaction-mapping table are also the structures whose cache-line contention ultimately bounds throughput at high core counts, a known scalability limit of SSN-style certification [7].

2.3 Write-Ahead Logging for Recovery

Durability and atomicity across crashes are provided by write-ahead logging (WAL). Under the WAL rule, a transaction’s log records are forced to stable storage before the transaction is reported as committed, so that a crash can be recovered by replaying the log. ARIES is the classical WAL design: each log record carries a monotonically increasing log sequence number (LSN), recovery repeats history in a redo pass and then rolls back loser transactions in an undo pass, and fuzzy checkpoints bound the amount of log that must be scanned [8], [9]. Forcing the log on every commit is costly because it requires a storage synchronization (*fdatsync*) on the commit path; group commit amortizes this cost by making a single synchronization durable for a

Algorithm 1 Per-Thread WAL Commit for Update Transactions (P-WAL)

```

1: Shared: global LSN counter  $L$ ; per-stream durable LSN  $f[i]$ ;
   global durable prefix  $P = \min_i f[i]$ 
2: function PWALCOMMIT( $T$ , worker  $w$ )
3:    $records \leftarrow \text{BUILDLOGRECORDS}(T)$      $\triangleright$  writes and commit
   record
4:   for all  $r \in records$  do
5:      $r.lsn \leftarrow \text{FETCHINC}(L)$              $\triangleright$  global LSN allocation
6:      $\text{APPEND}(\text{stream}[w], r)$                $\triangleright$  worker-private WAL stream
7:   end for
8:    $T.\ell \leftarrow \max\{r.lsn \mid r \in records\}$      $\triangleright$  commit LSN
9:    $\text{WRITE}(\text{stream}[w]); \text{FDATASYNC}(\text{stream}[w])$   $\triangleright$  worker waits
   for persistence
10:   $f[w] \leftarrow \max(f[w], T.\ell)$ 
11:   $P \leftarrow \min_i f[i]$                        $\triangleright$  global durable prefix
12:  while  $P < T.\ell$  do
13:     $P \leftarrow \min_i f[i]$ 
14:  end while
15:   $\text{ACKNOWLEDGE}(T)$ 
16: end function

```

batch of commits. In a main-memory database the redo log is the dominant persistence cost, so the rate at which log records can be made durable—and the latency of doing so—directly bounds commit throughput.

2.4 Parallel Logging

A single shared log serializes every commit through one log tail, which becomes a bottleneck on many cores. Parallel WAL designs remove this bottleneck by partitioning the log into multiple streams that are appended and flushed independently [10], [11], [12].

2.4.1 P-WAL

P-WAL gives each worker its own log stream, which we hereafter call a *WAL shard*, so that workers append commit and redo records without contending on a shared log tail [12]. Algorithm 1 shows the resulting commit path. Because the shards are independent, P-WAL re-establishes a single global order across them: every record is stamped with a value drawn from a global LSN counter L , and each shard publishes a durable position $f[i]$ whose minimum over all shards forms a global durable prefix $P = \min_i f[i]$. A transaction is acknowledged once the global prefix reaches its commit LSN, which guarantees that every earlier record, on every shard, is already durable. P-WAL underlies Shirakami, the transaction engine of Tsurugi [12], [13].

This design makes log insertion parallel, but combining it with a timestamp-based certifier such as SSN raises the three problems (P1–P3) detailed in Section 1.2.

3. Proposal: Ayame

Ayame is a logging method that integrates durable write-ahead logging into SSN while avoiding the three problems of P-WAL identified in Section 1.2. It combines three techniques. First, it reuses the commit timestamp that SSN already assigns as the logical log sequence number, eliminating the separate WAL-side global counter (addresses

P1). Second, it splits the commit processing into worker, flusher, and committer roles so that worker threads do not block on `fdatasync` or durable-ack waiting, subject only to a bounded backpressure limit (addresses **P2**). Third, it acknowledges a transaction using a *dependency-closed durable frontier* instead of a global durable prefix, so that a transaction waits only for the log records it actually depends on (addresses **P3**).

We use the following notation. A frontier F is a vector of per-shard log positions. $F \sqcup F'$ denotes the element-wise maximum, $F \leq F'$ holds iff $F[i] \leq F'[i]$ for every shard i , and $\perp$ denotes the empty frontier whose positions are all zero. D denotes the durable vector, where $D[i]$ is the largest log position of shard i that is known to be durable; each $D[i]$ advances monotonically.

3.1 Commit Timestamp as Logical LSN

In SSN, a committing transaction T is assigned a commit timestamp $T.c$ (its *cstamp*) that fixes its position in the serialization order. We assume that SSN assigns a unique commit timestamp to each committed update transaction, so these timestamps form a total order that recovery uses as the replay order. A separate WAL-side global LSN, as used by P-WAL, would redundantly re-encode the same logical order and add a second global-counter access to every commit. Ayame instead reuses $T.c$ directly as the logical LSN that orders recovery replay. No WAL-side global sequence number is allocated; in our evaluation the per-transaction WAL-side global-counter access drops to zero.

Removing the global *logical* LSN does not remove physical log addressing. Each WAL shard keeps a local position (a per-shard sequence number and byte offset) that locates a record within its log file. What Ayame eliminates is the duplicated global ordering counter, not the per-shard physical position.

3.2 Worker, Flusher, and Committer Separation

Ayame divides commit processing into three roles. A *worker* executes the transaction, validates it with SSN, assigns the commit timestamp, appends the log record to its own WAL shard, publishes durability metadata, and registers a durable-acknowledgment request. The worker then returns to its next transaction; it does not call `fdatasync` and does not wait for a durable acknowledgment. A *flusher* owns one WAL shard, batches the queued records, writes them, calls `fdatasync`, and advances the shard's durable position. A *committer* returns durable acknowledgments to clients once the durable condition of a registered transaction is met. Workers therefore do not block on storage synchronization or durable-ack waiting, and keep executing transactions at full concurrency even though every update commit must still be made durable; the cost of making the log durable is paid by the flusher and committer. The one exception is a bounded backpressure limit: to cap memory use and acknowledgment latency, a worker stalls only if the number of registered but not-yet-acknowledged durable

Algorithm 2 Worker-Side Commit Protocol (Ayame)

```

1: function COMMIT( $T$ )
2:    $T.c \leftarrow$  commit timestamp (cstamp) assigned by SSN       $\triangleright$ 
      logical LSN
3:   if SSNVALIDATE( $T$ ) fails then
4:     abort  $T$ ; return
5:   end if
6:   if  $T$ 's write set is empty then                                 $\triangleright$  read-only
7:     ACK( $T$ )                                                     $\triangleright$  read-only has no recoverable effect
8:     return logical read-only commit
9:   end if
10:   $s \leftarrow$  WAL shard assigned to this worker
11:   $T.seq \leftarrow$  RESERVESEQ( $wal[s]$ )     $\triangleright$  reserve this transaction's
      log position
12:   $F \leftarrow T.dep$ ;  $F[s] \leftarrow \max(F[s], T.seq)$      $\triangleright$  closed frontier
13:   $r \leftarrow$  BUILDRECORD( $T.c, F, T.write\_set$ )     $\triangleright$  commit record
      carries  $F$ 
14:  ENQUEUE( $wal[s], r$ )     $\triangleright$  append only; no flush, no wait
15:  for all versions  $v$  created by  $T$  do
16:     $v.wf \leftarrow F$ 
17:  end for
18:  PUBLISHVERSIONS( $T.write\_set$ )     $\triangleright$  visible only after  $wf$  is
      installed
19:  for all versions  $v$  read by  $T$  do
20:     $v.rf \leftarrow v.rf \sqcup F$                                  $\triangleright$  monotone merge
21:  end for
22:  REGISTER( $F, T$ )     $\triangleright$  durable ack deferred to the committer
23:  return after logical commit registration
24: end function
    
```

commits reaches a configured maximum. Accordingly, we report throughput as the number of durable acknowledgments returned to clients, not the number of logical commits.

Algorithm 2 shows the worker-side commit protocol. The commit timestamp $T.c$ assigned for SSN validation doubles as the logical LSN (line 2). After validation, the worker reserves its log position on its shard (line 11) and forms the closed frontier F by extending its dependency frontier $T.dep$ with that position (line 12). It then builds the log record, whose commit record carries F (line 13), and enqueues it without flushing or waiting (line 14). Ayame installs F as the write frontier of the versions T created (line 16) and publishes those versions (line 18); a version becomes visible only after its write frontier is installed, so any later reader or overwriter observes the writer's complete frontier. It also merges F into the read frontiers of the versions T read (line 20). Finally, the worker registers F for durable acknowledgment (line 22) and returns to the execution scheduler; the client-visible commit response is returned only later by the committer, once the frontier condition is satisfied.

The flusher does not require an algorithm of its own. Each flusher owns one WAL shard, repeatedly collects the records queued by workers, writes them as a single batch, calls `fdatasync` once for the batch, and then advances the shard's durable position by invoking `ONDURABLEADVANCE` (Algorithm 4). Batching many records per `fdatasync` is what raises the number of commits made durable per synchronization and reduces flush pressure relative to P-WAL's

Algorithm 3 Dependency Collection (Ayame)

```

1: function BEGIN( $T$ )
2:    $T.dep \leftarrow \perp$ 
3: end function

4: function READVERSION( $T, v$ )
5:   add  $v$  to  $T$ 's read set
6:   if  $T$  is known read-only then
7:     return ▷ SI read-only fast path
8:   end if
9:    $T.dep \leftarrow T.dep \sqcup v.wf$  ▷ the writer of  $v$ 
10: end function

11: function OVERWRITEVERSION( $T, v$ )
12:   add the new version to  $T$ 's write set
13:    $T.dep \leftarrow T.dep \sqcup v.wf \sqcup v.rf$  ▷ the writer of  $v$  and its readers
14: end function

```

per-transaction synchronization.

3.3 Dependency-Closed Durable Acknowledgment

The remaining question is when a logically committed transaction may be acknowledged as durable. Ayame answers this with a frontier attached to each version. A version carries a *write frontier* wf , the closed frontier of the transaction that created it. A writer publishes wf before its version becomes visible, so any transaction that later reads or overwrites the version observes the complete write frontier of its producer. During execution, a transaction joins these write frontiers into its dependency frontier $T.dep$, as shown in Algorithm 3: reading a version makes T depend on the writer of that version (line 9), and overwriting a version makes T depend on the writer of the version it supersedes (line 13). A version also carries a *read frontier* rf , into which readers monotonically merge their closed frontiers; OVERWRITEVERSION additionally joins rf so that an overwriter conservatively tracks the readers of the version it supersedes. This read-frontier term is a conservative delay mechanism; it is not required for the recoverable-acknowledgment proof, which relies only on write frontiers published before a version becomes visible. Transactions known to be read-only skip WAL generation and durable-acknowledgment registration. When the read-only fast path applies, they also avoid dependency-frontier collection; otherwise, only lightweight read-path frontier bookkeeping may remain.

At commit, the closed frontier F (line 12 of Algorithm 2) extends $T.dep$ with T 's own log position. By induction over the dependency order, F therefore contains the log positions of the transitive closure of the read-from and version-order (write-write) dependencies of T , all of which are recorded through write frontiers. Publishing F as the write frontier of the versions written by T lets future readers and overwriters inherit T 's closed frontier through Algorithm 3. Merging F into the read frontier of the versions read by T conservatively delays future overwriters of those versions, but this

Algorithm 4 Durable Acknowledgment (Ayame)

```

State: durable vector  $D$ ; per-shard waitlists  $W[i]$  ordered by required position

1: function REGISTER( $F, T$ ) ▷ atomic w.r.t.
   ONDURABLEADVANCE
2:    $T.remaining \leftarrow 0$ 
3:   for all shards  $i$  such that  $F[i] > D[i]$  do
4:      $T.remaining \leftarrow T.remaining + 1$ 
5:     insert  $(F[i], T)$  into  $W[i]$ 
6:   end for
7:   if  $T.remaining = 0$  then
8:     ACK( $T$ ) ▷ own record and all dependencies durable
9:   end if
10: end function

11: function ONDURABLEADVANCE( $i, d$ ) ▷ called by flusher  $i$  after fdatasync
12:    $D[i] \leftarrow d$ 
13:   while  $W[i]$  is not empty and  $\text{MINKEY}(W[i]) \leq d$  do
14:      $(need, T) \leftarrow \text{POPMIN}(W[i])$ 
15:      $T.remaining \leftarrow T.remaining - 1$ 
16:     if  $T.remaining = 0$  then
17:       ACK( $T$ )
18:     end if
19:   end while
20: end function

```

read-frontier term is not required for the safety proof.

Algorithm 4 shows the committer side. REGISTER parks T on each shard whose durable position has not yet reached the requirement of F (line 5); if no shard is behind, T is acknowledged immediately (line 8). ONDURABLEADVANCE, invoked by a flusher after **fdatasync**, advances the durable position of one shard and wakes the parked transactions whose requirement on that shard is now satisfied, releasing the acknowledgment once a transaction's last outstanding shard becomes durable (line 17). REGISTER and ONDURABLEADVANCE execute under a single lock so that a durable advance cannot slip between the durability test and the parking step.

The acknowledgment rule is exactly $F \leq D$: a transaction is acknowledged once its own record and every record it transitively depends on are durable. We now make the safety of this rule precise. For a committed update transaction U , let s_U and $U.seq$ be the shard and position of its log record, so its published closed frontier F_U satisfies $F_U[s_U] \geq U.seq$. Let $\text{cl}(T)$ be the set of update transactions reachable from T through read-from dependencies (T reads a version written by U) and version-order dependencies (T overwrites a version written by U), including T itself. Both kinds of dependency are recorded by joining the relevant version's write frontier, which its writer publishes before the version becomes visible.

Lemma 1 (Dependency-closed durability) If ACK(T) is invoked, then for every update transaction $U \in \text{cl}(T)$ the WAL record of U is durable.

Proof. Dependency collection (Algorithm 3) joins the write frontier of each version T reads or overwrites into $T.dep$, and

the commit protocol sets $F = T.dep$ with $F[s_T] \geq T.seq$ (line 12). A writer publishes its write frontier before its version becomes visible, so T observes the complete frontier F_U of every U it directly depends on; since each F_U is itself closed, induction over the dependency order gives $F \geq F_U$, hence $F[s_U] \geq F_U[s_U] \geq U.seq$, for every $U \in cl(T)$. $ACK(T)$ is invoked only when $F \leq D$ (Algorithm 4), so $D[s_U] \geq U.seq$. Since $D[i]$ is a durable prefix of shard i , the record of U is durable.

Each commit record stores the transaction's commit timestamp $T.c$ and its closed dependency frontier F , alongside the redo records that carry the write set. During recovery, Ayame reconstructs the durable vector D by parsing each shard's records in order and advancing $D[i]$ until the first record that does not parse as a complete, well-formed entry, which indicates a torn final write; records beyond that point are ignored. A distinguished commit marker identifies each transaction's commit record. Ayame then collects the committed transactions whose commit records lie within D , keeps only those whose stored frontier is covered by the durable vector (that is, $F \leq D$), and replays this dependency-closed set in increasing $cstamp$ order. Because the replayed set is closed under $F \leq D$, a recovered transaction never appears without the transactions it depends on, even if its own commit record happened to be durable while a dependency's was not.

Theorem 1 (Recoverable acknowledgment)

Suppose a crash occurs and recovery replays exactly the transactions whose stored frontier is covered by the durable vector ($F \leq D$), in increasing $cstamp$ order. If T was acknowledged before the crash, recovery never yields a state that contains the effects of T but omits the effects of some $U \in cl(T)$.

Proof. By Lemma 1, every $U \in cl(T)$ has a durable WAL record. Moreover, the induction in that lemma gives $F \geq F_U$ for every $U \in cl(T)$, where F is the frontier T registered; since $ACK(T)$ requires $F \leq D$, it follows that $F_U \leq D$. Every $U \in cl(T)$ therefore belongs to the dependency-closed set that recovery admits—the committed transactions whose stored frontier is covered by D —not merely to the set of physically durable records, and is replayed. For the dependency classes included in $cl(T)$ —read-from dependencies and version-order dependencies from overwritten versions—the producer version must have become visible before T reads or overwrites it, so under the engine's MVCC visibility rule the producer's commit timestamp precedes T 's. Because logical LSNs equal commit timestamps, replaying the acknowledged transactions in increasing $cstamp$ order installs every such U before T . Hence whenever T 's effects are installed, the effects of all $U \in cl(T)$ are already present.

The lemma and theorem rest only on write frontiers, which are published before a version is visible and are therefore observed completely by every reader and overwriter. The read-frontier term that `OVERWRITEVERSION` joins (the rf of an overwritten version) makes acknowledgment additionally wait for the readers of that version,

an anti-dependency. This term is a conservative over-approximation: it can only delay an acknowledgment, never release one early, and the safety argument above does not use it. Because readers publish rf concurrently with overwriters, it is a best-effort addition rather than a guaranteed bound, which is why we keep the proof over write frontiers alone.

The rule is also less conservative than a global durable prefix: a transaction waits only on the shards that appear in its frontier, so an unrelated slow shard never delays it.

4. Evaluation

4.1 Implementation

We implement Single WAL, P-WAL, and Ayame on CCBench [7], a concurrency-control benchmarking framework with minimal functionality for protocol analysis, using Masstree as the index and `mimalloc` as the allocator. The code is written in C++20 and compiled with GCC at the `-O3` optimization level. Our implementation is publicly available.*¹

A key methodological point is that Single WAL, P-WAL, and Ayame are implemented in the *same* binary and selected at run time by environment flags (`CCBENCH_WAL_MODE` and `CCBENCH_WAL_DURABLE_MODE`). The CCBench SI+SSN protocol implementation, the index, the workload generator, and the pre-commit path are therefore identical across the three systems; only the WAL durability path differs. Single WAL appends to one shared log under a global latch and synchronizes it per commit. P-WAL gives each worker a private WAL shard and synchronizes it per commit, acknowledging through a global durable prefix. Ayame uses the worker/flusher/commmitter pipeline of Section 3 with dependency-closed acknowledgment. Ayame keeps one WAL shard per flusher, so the number of shards equals the flusher count (nine at 48 workers) and is smaller than the worker count; each worker is mapped to a shard and several workers share one, which lets a flusher batch many workers' records into a single `fdatasync` (in contrast to P-WAL, which gives every worker a private shard). The dependency frontier is thus a vector over these shards, and the per-shard sequence reservation is shared by co-located workers. Each WAL redo record carries the key, value, storage identifier, and operation type of a write; the terminating commit record additionally stores the transaction's commit timestamp and its closed dependency frontier, which recovery uses to admit only the dependency-closed set of durable transactions. Read-only WAL skipping is enabled for all three systems; once a transaction is known to be read-only it produces no WAL record. For read-only transactions Ayame also skips durable-acknowledgment registration, but depending on the read path it may still execute lightweight frontier bookkeeping; YCSB-C quantifies this residual overhead.

Following the worker/flusher/commmitter separation of

*¹ The source code is available at <https://github.com/sa2shun/ccbench-wal/tree/cstamp-pwal-framework>.

Table 1 Default parameter settings.

Type	Parameter	Setting
DB	#records	100,000
	value size	4B
	key distribution	uniform
TX	#operations	10
	read ratio	50 / 95 / 100 %
Run	worker threads	1,12,24,36,48,60,72,84,96
	measurement time	5 s
	repeats	5
Ayame	flush group size [commits]	256
	flush interval	50 μ s
	max pending acks	65,536

Section 3, Ayame offloads durability to background threads: it uses 1, 3, 5, 7, 9, 12, 14, 17, and 19 flusher threads for 1, 12, 24, 36, 48, 60, 72, 84, and 96 worker threads, respectively, and one committer thread in all configurations. The x-axis of our worker-scaling experiments is the number of transaction worker threads; at 48 workers Ayame therefore runs 58 active threads (48 workers, 9 flushers, and one committer). At 96 workers Ayame runs 116 active threads (96 workers, 19 flushers, and one committer), which exceeds the 96 logical cores of the machine; we revisit this oversubscription when discussing the high-thread end of the sweep.

4.2 Experimental Configuration

4.2.1 Environment

Experiments were conducted on a server with two sockets of Intel(R) Xeon(R) Gold 5418N CPUs. Each socket had 24 physical cores with two hardware threads per core, for 48 physical and 96 logical cores in total, organized as two NUMA nodes. Each core had a 48KiB private L1d cache and a 2MiB private L2 cache, and each socket shared a 45MiB L3 cache. The DRAM capacity was 256 GB. The machine ran Ubuntu 22.04.5 LTS (Linux kernel 5.15), and WAL files were placed on a locally attached SSD formatted with ext4. Worker threads were pinned to distinct logical cores with `sched_setaffinity`.

4.2.2 Benchmark and Workloads

All experiments used the Yahoo! Cloud Serving Benchmark (YCSB) [14]. We evaluated three representative workloads: YCSB-A (50% read, 50% update), YCSB-B (95% read, 5% update), and YCSB-C (read-only). These workloads exercise the WAL along a spectrum from update-heavy to read-only and are widely used to evaluate concurrency-control protocols. Each transaction issued a fixed number of ten operations over a database of 100,000 records with 4-byte values, and keys were drawn from a uniform distribution.

4.2.3 Metric

The primary metric is *durable-acknowledgment throughput*: the number of transactions acknowledged as durable to the client, divided by the measurement time. All systems are measured at the same stop barrier, and acknowledgments drained during shutdown are excluded. For the worker-wait systems (Single WAL and P-WAL) a logical commit and a durable acknowledgment coincide, so the met-

ric has the same meaning across the three systems. We additionally report the p99 durable acknowledgment latency and the number of pending (not-yet-acknowledged) durable commits. Unless otherwise stated, each reported value is the mean of the corresponding per-run metric over five runs; reported p99 latencies are means of the per-run p99 values.

4.3 Results

This section evaluates Ayame as a complete durable-acknowledgment pipeline. The results show that Ayame substantially improves end-to-end throughput on update workloads by decoupling workers from batched durable flushing, while keeping tail latency and pending durable commits small. The remaining throughput decline at high thread counts is caused by contention in the underlying SSN concurrency-control layer rather than by the WAL pipeline.

4.3.1 End-to-End Throughput Improves on Update Workloads

We evaluate Ayame using the worker-thread sweep of Table 1, which adds a single-thread baseline and then spans 12 to 96 worker threads in steps of 12 (up to the full two-socket machine). Figure 2 plots the durable-acknowledgment throughput across the sweep, and Table 2 lists the 48-worker numbers. At 48 worker threads on YCSB-A, Single WAL achieves 9K durable acknowledgments per second and P-WAL achieves 50K, whereas Ayame achieves 253K; on YCSB-B, Single WAL achieves 26K and P-WAL 165K, whereas Ayame achieves 467K. These are $5.1\times$ and $2.8\times$ improvements over P-WAL on YCSB-A and YCSB-B, and the Ayame-over-P-WAL ratio stays large across the sweep, ranging from $6.5\times$ at 12 workers to $4.1\times$ at 96.

This gain comes from Ayame’s durability pipeline as a whole, mainly from decoupling workers from batched durable flushing (A2); reusing the commit timestamp as the logical LSN (A1) is a design simplification that removes duplicated logical ordering rather than a direct throughput optimization. This end-to-end comparison should therefore not be read as isolating the effect of dependency-closed acknowledgment (A3), whose contribution we isolate in Section 4.4.

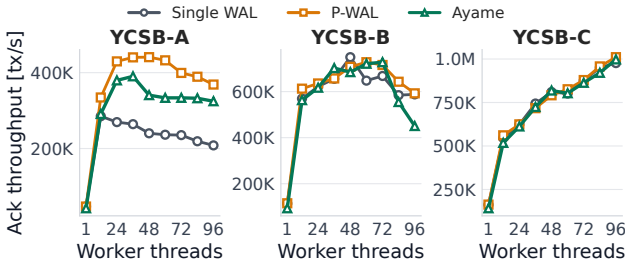
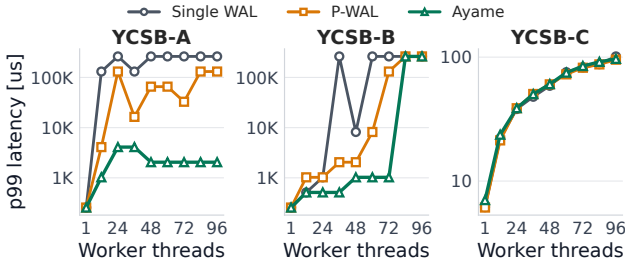
We also evaluate YCSB-C, where all transactions are read-only and WAL records are skipped. Durability work largely disappears: with the same binary and a read-only empty-frontier fast path, the three systems land within about 8% of one another at 48 worker threads (872K for Single WAL, 808K for P-WAL, and 812K for Ayame in Table 2); Ayame is marginally below Single WAL because it may still execute lightweight read-path frontier bookkeeping. We therefore use YCSB-C as a sanity check for read-only bookkeeping overhead rather than as evidence of WAL persistence performance.

Table 2 summarizes the main 48-worker end-to-end results. For YCSB-C, Tx/sync is not applicable because no WAL records are flushed.

Table 2 End-to-end YCSB results at 48 transaction worker threads. Arrows mark the better direction and bold marks where Ayame wins.

Workload	System	Ack tps $\uparrow$	p99 μ s $\downarrow$	Pending $\downarrow$	Tx/sync $\uparrow$	WAL atomic/tx $\downarrow$
YCSB-A	Single WAL	8.8K	8,192	0	1.0	6.01
	P-WAL	49.8K	1,024	0	1.0	6.00
	Ayame	252.7K	2,048	318	25.4	0.00
YCSB-B	Single WAL	25.7K	4,096	0	2.5	0.90
	P-WAL	165.5K	512	0	2.5	0.90
	Ayame	466.9K	512	83	16.7	0.00
YCSB-C	Single WAL	871.6K	53	0	—	0.00
	P-WAL	807.5K	58	0	—	0.00
	Ayame	811.7K	57	0	—	0.00

Ack tps is durable-acknowledgment throughput; Tx/sync is commits made durable per `fdatasync`; WAL atomic/tx is per-transaction WAL-side global-counter accesses. Ayame uses 9 flusher threads and 1 committer at 48 workers.

**Fig. 2** Durable-acknowledgment throughput by transaction worker threads. Ayame improves update and mixed read-write workloads through its complete durability pipeline, mainly by decoupling workers from batched durable flushing. The effect of dependency-closed acknowledgment is isolated in Figure 5.**Fig. 3** P99 durable-acknowledgment latency by transaction worker threads.

4.3.2 Tail Latency and Pending Durable Commits Remain Small

Ayame keeps pending durable commits small (Table 2): at 48 worker threads it holds 318 pending commits with a 2 ms p99 acknowledgment latency on YCSB-A, and 83 with a 0.5 ms p99 on YCSB-B. Figure 3 shows that this p99 stays low across the whole sweep. The asynchronous pipeline therefore keeps up with the offered durable-logging load at every thread count; it does not buy throughput by letting an unbounded backlog accumulate. The zero pending count of Single WAL and P-WAL does not mean that they avoid waiting; their workers wait synchronously on the commit path, so the wait shows up as lower throughput rather than as an asynchronous pending queue.

4.3.3 The Scalability Ceiling Comes from SSN Contention

The high-thread throughput decline is a known scalability limit of the underlying SSN concurrency control, not a

limit of Ayame’s WAL pipeline: SSN certifies serializability through centralized metadata—a single global commit-timestamp counter (which Ayame reuses as the logical LSN, **A1**) and a central transaction-mapping table—and this metadata does not scale to many cores [7]. Concretely, Ayame’s YCSB-A throughput peaks near 24–36 workers (294–296K) and then declines toward 96 workers (209K); YCSB-B follows the same shape, and P-WAL declines in parallel. Figure 4 traces the cause to per-transaction cost rather than to aborts. The cycles spent per committed transaction rise steeply with thread count—about $2.8\times$ on YCSB-A (136K at 12 workers to 375K at 96) and $2.2\times$ on YCSB-B—consistent with cache-line contention on these shared structures as they bounce between cores’ caches; because the per-transaction cost grows faster than the core count, throughput falls once the contention outweighs the added parallelism. Aborts, by contrast, cannot explain the decline: the abort rate never exceeds 1.5% on YCSB-A and 0.03% on YCSB-B, and YCSB-B shows the same decline with that near-zero abort rate. Nor is the decline thread oversubscription: P-WAL carries no background flush threads, so at 96 workers it runs 96 threads on the 96 logical cores without oversubscribing them, yet its throughput falls in parallel with Ayame’s. Finally, the limit is upstream of durability—the p99 acknowledgment latency stays at or below 2 ms across the whole sweep (Figure 3) and the pending durable commits remain small (Table 2)—so the flushers and committer are not the bottleneck; the ceiling is set by the concurrency-control layer that Ayame inherits through cstamp reuse.

4.4 Acknowledgment-Policy Comparison

This section isolates the effect of dependency-closed acknowledgment (**A3**). Both variants use the same asynchronous worker/flusher/committer pipeline and differ only in the acknowledgment condition: the *global-prefix* variant acknowledges a transaction only after a single global durable prefix covers its logical commit position, whereas the *dependency-frontier* variant acknowledges it once the shard positions in its dependency frontier are durable. The results show that **A3** trades some short-window throughput for bounded pending durable commits and much lower tail latency.

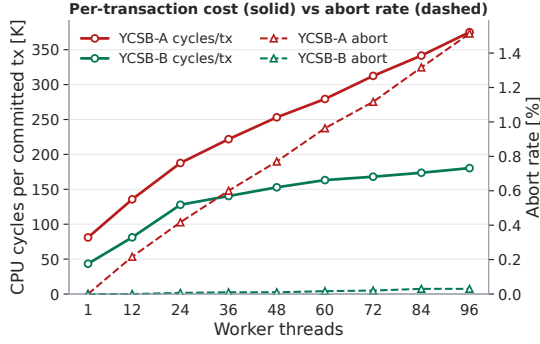


Fig. 4 Per-committed-transaction CPU cost (solid, left axis) and abort rate (dashed, right axis) versus worker threads for Ayame. The rise in cycles per committed transaction is consistent with cache-line contention on SSN’s central commit-timestamp counter and transaction-mapping table, while the abort rate stays small; YCSB-B declines with a near-zero abort rate.

4.4.1 A3 Bounds Backlog and Tail Latency, Not Raw Throughput

The benefit of dependency-closed acknowledgment is not that a transaction has many dependencies; it appears when a transaction does *not* depend on some slow or lagging WAL shards, because global-prefix acknowledgment still waits for those shards whereas the dependency frontier does not.

Figure 5 reports the result. On YCSB-A the contrast is sharp: once multiple WAL shards are active, the global-prefix variant stays pinned near the 65,536 backpressure limit with a 131 ms p99 acknowledgment latency, whereas the dependency-frontier variant remains below 500 pending durable commits—far below the global-prefix queue—with a 1–4 ms p99. At 48 workers, for instance, global-prefix holds 64,197 pending commits and 131 ms while Ayame holds 455 and 2 ms. Even without injecting an artificial straggler, natural progress variation among the WAL shards is enough to keep the global durable prefix behind, so the global-prefix queue never drains; the dependency-frontier rule does not wait for unrelated shards and therefore drains regardless of worker count. On YCSB-B, whose update load is an order of magnitude lighter, the dependency-frontier rule again holds the pending count under about 100 and the p99 at 1 ms across the whole sweep, while the global-prefix rule drains at low-to-mid worker counts but saturates beyond 48 workers as shard progress fluctuates. (Pending counts here are the measurement-window snapshot, as in Table 2; these acknowledgment-policy runs are separate from the main worker-scaling runs, so values differ slightly due to run-to-run variation.) This comparison also decomposes the end-to-end gain. Moving from P-WAL’s worker-wait commit to Ayame’s asynchronous worker/flusher/committer pipeline (techniques **A1** and **A2**), while still acknowledging at the global durable prefix, raises YCSB-A throughput at 48 workers from 50K to 444K durable acknowledgments per second—about 8.9 $\times$, and the source of Ayame’s throughput advantage over P-WAL. Switching the acknowledgment rule of that same pipeline from global-prefix to the dependency frontier (**A3**) is *not* a throughput optimization in this

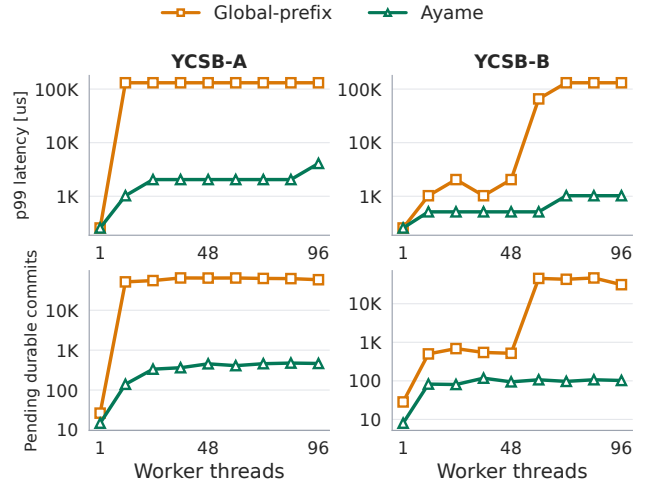


Fig. 5 Acknowledgment-policy comparison (top: p99 latency; bottom: pending durable commits). Both variants use the same asynchronous WAL pipeline and differ only in the acknowledgment condition. The global-prefix rule saturates on YCSB-A once multiple WAL shards are active, while the dependency-frontier rule keeps pending commits and p99 acknowledgment latency bounded.

short measurement window: it lowers throughput to 251K. The global-prefix variant can show higher short-window acknowledgment throughput because it accumulates a large backlog and releases many acknowledgments in bulk when the prefix advances, at the cost of a saturated pending queue and high tail latency. Dependency-closed acknowledgment instead bounds the backlog and tail latency, cutting pending durable commits from 64,197 to 455 and the p99 from 131 ms to 2 ms. Because the two variants share the same pipeline and differ only in the acknowledgment condition, this comparison isolates the effect of **A3**.

4.4.2 Avoiding Unrelated-Shard Waits Helps Even with Sparse Frontiers

The benefit comes from not waiting for shards a transaction does not depend on, so it should not require dense dependency frontiers. Zipfian skew does not let us vary frontier density: it changes which keys are hot but not which WAL shards wrote them, so in a non-partitioned workload the writers of the accessed versions stay spread across all shards and the dependency frontier remains dense. We therefore vary the frontier width directly with an abort-free partitioned workload in which each worker owns a key partition and a remote-access probability controls how often a transaction touches other partitions. With `remote = 0` every transaction depends only on its local shard—a sparse frontier of one shard per transaction ($nz/tx = 1$)—and with remote accesses the frontier quickly becomes dense (about nine shards). The jump is abrupt because dependency frontiers compose transitively—once a transaction touches a remote partition it inherits that partition’s own frontier—so even a 1% remote-access probability pulls in almost all shards. Because this abort-free workload commits—and therefore logs—on every transaction, it is the most write-intensive workload in our evaluation; the flush group of 256 commits

Table 3 Dependency-density experiment at 48 worker threads on an abort-free partitioned workload. The remote-access probability sets the frontier width (nz/tx is the mean number of shards per dependency frontier). Arrows mark the better direction and bold marks where Ayame wins.

Remote	nz/tx	Policy	Ack tps ↑	p99 ms ↓	Pending ↓
0%	1.0	Global	438.9K	131	64,626
		Ayame	261.7K	2	397
1%	8.9	Global	437.3K	131	63,618
		Ayame	256.8K	2	325
10%	8.9	Global	436.6K	131	64,602
		Ayame	252.8K	2	358
100%	9.0	Global	438.9K	131	64,924
		Ayame	238.5K	2	349

(Table 1) keeps the flush pipeline unsaturated for *both* acknowledgment policies, so the remaining difference reflects only the acknowledgment rule rather than flush-pipeline saturation.

Table 3 reports the result. The global-prefix rule essentially saturates the pending-ack queue and holds a 131 ms p99 at every density, including the fully sparse case, because it waits for the global durable prefix regardless of what a transaction actually depends on. The dependency-frontier rule keeps the pending count to a few hundred commits and the p99 to about 2 ms across the whole range; its throughput is again slightly lower, as in the ablation. Crucially, even when each transaction depends on a single shard (nz/tx = 1), the dependency-frontier rule still keeps pending durable commits and tail latency far below the global-prefix rule: about 400 versus 65,000 pending commits, and 2 ms versus 131 ms p99. The value of dependency-closed acknowledgment is therefore not that a transaction has many dependencies, but that it does not wait for the shards it does not depend on, and this value remains when frontiers are sparse.

4.5 Hardware Counters and Overhead

Table 4 reports a perf-stat comparison of the three systems at 48 worker threads, showing two sources of Ayame’s throughput improvement and one overhead. First, Ayame amortizes storage synchronization: it makes 23 commits durable per `fdatasync` on YCSB-A and 14 on YCSB-B, whereas Single WAL and P-WAL synchronize roughly once per transaction (Tx/sync of 1.0–2.5). Second, Ayame spends far fewer CPU cycles per committed transaction than P-WAL (243K versus 1.15M on YCSB-A): in P-WAL a worker cannot proceed until its own commit is durable, so worker progress is coupled to storage synchronization, whereas Ayame’s workers enqueue a log record and return to execution. Single WAL, in contrast, barely uses the machine (1.8 cores) because its workers serialize on a single shared log. Third, this decoupling is paid for in context switches, of which Ayame performs the most, reflecting the worker/flusher/commmitter hand-offs. On read-only YCSB-C, the three systems converge in throughput, core utilization, and cycles per transaction; Ayame still incurs more context switches due to the background pipeline, but this

does not materially affect read-only throughput. These perf-stat runs are separate from the main throughput runs and carry profiling overhead, so their absolute throughputs are lower (for instance, YCSB-C reads about 683K here versus 812K in Table 2); we use them only for the relative cross-system comparison.

5. Related Work

As the number of CPU cores has scaled up in recent years, the traditional design of utilizing a single log buffer and a single log flusher has increasingly become a critical performance bottleneck, primarily due to severe contention over the log stream. To mitigate this issue, extensive research has been conducted on parallel logging or decentralized logging, where multiple cores record and append log records to persistent storage concurrently.

Pioneering efforts in parallel logging include systems such as Ather [?] and passive group commit [?]. These approaches allocate isolated log buffers (partitioned logs) per core or per socket, thereby completely eliminating latch contention during log insertion. However, a naive partitioning of logs introduces a fundamental trade-off: it complicates the precise reconstruction of global dependencies—such as the exact commit order of transactions—during crash recovery. To manage these dependencies efficiently, SiloR [10] introduced a cooperative, epoch-based mechanism that flushes the log records of all cores collectively at fixed time intervals (epochs), ensuring consistency at the granularity of an epoch. Meanwhile, Taurus [15] embeds logical timestamps into log records to capture data dependencies among transactions. By dynamically reconstructing a dependency graph during the recovery phase, Taurus successfully achieves both out-of-order parallel log writes and accurate recovery.

Furthermore, the emergence of non-volatile RAM (NVRAM) has brought a paradigm shift to the design of parallel logging protocols. Because NVRAM offers byte-addressability and low latency, it challenges the fundamental premise of traditional Write-Ahead Logging (WAL), which relies on translating in-memory data modifications into a serialized, disk-oriented log stream.

For instance, Write-Behind Logging (WBL) [16] leverages the persistence of NVRAM by flushing the modified data itself directly to the persistent layer before a transaction commits. By recording only minimal commit metadata in the log, WBL drastically reduces the overhead of generating Redo logs. Other studies, such as Rethinking Logging for NVRAM [?] or similar fine-grained approaches, propose parallel, fine-grained checkpointing on NVRAM, which either eliminates the need for operational logs entirely or significantly restricts their role. Additionally, systems like XXX [?] allocate independent, per-core log spaces directly within NVRAM, enabling each core to perform synchronous writes without cross-core coordination, thereby achieving parallel logging throughput that approaches the physical limits of the hardware.

In summary, modern logging mechanisms continue to

Table 4 Perf-stat summary at 48 transaction worker threads. Cores is CPU-core utilization, Cyc/tx is CPU cycles per committed transaction, Ctx sw. is total context switches, and Tx/sync is commits per `fdatasync`. Arrows mark the better direction and bold marks where Ayame wins.

Workload	System	Ack tps ↑	Cores	Cyc/tx ↓	Ctx sw	Tx/sync ↑
YCSB-A	Single WAL	8.1K	1.8	552K	133K	1.0
	P-WAL	48.7K	22.1	1.15M	1.48M	1.0
	Ayame	237.4K	23.5	243K	4.21M	23.0
YCSB-B	Single WAL	22.1K	1.4	165K	140K	2.5
	P-WAL	162.6K	18.0	282K	1.29M	2.5
	Ayame	414.4K	24.3	145K	4.12M	14.4
YCSB-C	Single WAL	680.6K	48.4	184K	4K	—
	P-WAL	685.5K	48.3	183K	4K	—
	Ayame	683.0K	48.4	184K	34K	—

evolve toward a harmonious integration of multi-core parallelism and the byte-addressable persistence provided by NVRAM.

6. Conclusion

This paper presented Ayame, a timestamp-integrated, dependency-closed parallel WAL protocol for SSN-based main-memory transaction processing. Ayame reuses SSN's commit timestamp as the logical LSN (**A1**), separates transaction execution from durable acknowledgment across worker, flusher, and committer roles (**A2**), and acknowledges each transaction once its own log record and its dependency frontier are durable (**A3**). The throughput improvement—a 4.1- to 6.5-fold gain over P-WAL on YCSB-A across the worker sweep—comes mainly from the worker/flusher/committer pipeline and batched durable flushing (**A2**), while timestamp integration (**A1**) removes redundant WAL-side logical ordering. Dependency-closed acknowledgment (**A3**), in turn, bounds the durable-acknowledgment backlog and tail latency without waiting for unrelated WAL shards, which we proved recoverable (Lemma 1, Theorem 1).

The scope of this paper is the durable-acknowledgment path and its safety: we evaluated durable-acknowledgment throughput, pending durable commits, and tail latency, and we established that an acknowledged transaction is always recoverable together with the transactions it depends on. We did not measure crash-recovery time or replay throughput; quantifying recovery cost, and evaluating Ayame under a wider range of workloads and storage devices, are left as future work.

Acknowledgments

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